PROJECT LIST (Just for the IDEAS)

Project 1: 6G Integrated Sensing and Communication (ISAC) Front-End Co-Design

- **The Research Gap:** In 6G, devices will use the same RF waveform for both data communication (WiFi/5G) and radar sensing (gesture recognition, automotive). However, waveforms optimized for radar (like FMCW) have terrible Peak-to-Average Power Ratios (PAPR) that destroy Power Amplifier (PA) efficiency, while communication waveforms (OFDM) have poor radar ambiguity.
- **The Task:**
    1. Design a hybrid waveform (e.g., OFDM-FMCW) in MATLAB.
    2. Model a realistic PA (using Keysight ADS) and analyze how the hybrid waveform degrades the Error Vector Magnitude (EVM) for comms and the Side-Lobe Level (SLL) for radar.
    3. Propose a novel Digital Pre-Distortion (DPD) or PA back-off optimization technique that balances both.
- **Target Paper Title:** *"Joint Waveform and Power Amplifier Back-off Optimization for 6G Integrated Sensing and Communication Systems."*
- **Target Venue:** *IEEE Radar Conference* or *IEEE Wireless Communications Letters*.

Project 2: Reconfigurable Intelligent Surfaces (RIS) for Spatial EMC and Interference Mitigation

- **The Research Gap:** Reconfigurable Intelligent Surfaces (metasurfaces) are typically used just to "boost" signals. A novel research angle is using them for **spatial EMC**—dynamically creating "spatial nulls" to block jammers or protect sensitive receivers from interference in dense urban environments.
- **The Task:**
    1. Design a tunable metasurface unit cell (using varactor diodes) in HFSS/CST.
    2. Extract the S-parameters and build an array factor model in MATLAB/ADS.
    3. Develop an algorithm that dynamically steers a beam to a legitimate user while simultaneously placing a deep "null" (EMC concept) in the direction of an interfering source.
- **Target Paper Title:** *"Dynamic Spatial Null-Steering and Interference Mitigation using Reconfigurable Intelligent Surfaces for mmWave EMC."*
- **Target Venue:** *IEEE Transactions on Antennas and Propagation* or *IEEE International Symposium on Electromagnetic Compatibility (EMC)*.

Project 3: Machine Learning-Assisted EMC/EMI Surrogate Modeling for Antenna-in-Package (AiP)

- **The Research Gap:** As 5G moves to mmWave (28 GHz, 39 GHz), antennas and RF ICs are integrated into the same tiny package (AiP). Simulating full-wave 3D EM crosstalk between these dense components takes weeks of compute time. Using Machine Learning to predict EM coupling (surrogate modeling) is a massive, highly-citable research trend.

- **The Task:**
    1. Generate a dataset of S-parameters (S21*S*21 coupling) between closely spaced mmWave traces/antennas using a script in HFSS/CST (varying trace width, spacing, and via fencing).
    2. Train a simple Neural Network or Gaussian Process model in Python to predict the EM coupling instantly based on geometry.
    3. Prove that the AI model predicts EMC violations 1000x faster than full-wave simulation with >95% accuracy.
- **Target Paper Title:** *"Deep Learning-Based Surrogate Modeling for Rapid EMC/EMI Crosstalk Prediction in mmWave Antenna-in-Package Systems."*
- **Target Venue:** *IEEE Transactions on Microwave Theory and Techniques (TMTT)* or *IEEE MTT-S International Microwave Symposium (IMS)*.

Project 4: Wideband SWIPT (Simultaneous Wireless Information and Power Transfer) Receiver Architecture

- **The Research Gap:** IoT sensors need to harvest RF energy from the environment while simultaneously decoding data. The challenge is that the non-linear diodes used for energy harvesting severely distort the communication signal. Designing a front-end that splits the signal and optimizes *both* the harvester's RF-to-DC efficiency and the receiver's sensitivity is a classic, highly publishable microwave problem.
- **The Task:**
    1. Design a novel RF power splitting network (e.g., a tunable diplexer) in ADS.
    2. Analyze the non-linear trade-off: if you send 80% of the power to the energy harvester, how much does the Bit Error Rate (BER) of the communication link degrade?
    3. Propose a dynamic matching network that adapts based on the incoming RF power level.
- **Target Paper Title:** *"Adaptive RF Front-End Architecture for Wideband Simultaneous Wireless Information and Power Transfer (SWIPT) in IoT Networks."*
- **Target Venue:** *IEEE Microwave and Wireless Technology Letters (MWTL)* or *IEEE Radio Wireless Week (RWW)*.

Project 5: Coexistence Filter Design for Dense IoT/Wi-Fi Environments

- **The Concept:** Modern IoT devices pack Wi-Fi, Bluetooth, and 5G into a tiny PCB. The Wi-Fi transmitter often blinds the Bluetooth receiver (a classic EMC/EMI desensitization problem). Students will design a highly compact, high-rejection "coexistence filter" (diplexer or bandstop) using Defected Ground Structures (DGS) or Substrate Integrated Waveguides (SIW).
- **The Task:**
    1. Design a microstrip bandpass filter with a deep notch (rejection) at a specific interfering frequency (e.g., passing 2.4 GHz Wi-Fi but creating a sharp null at 5 GHz).

2. Implement the filter using DGS (etching slots in the ground plane) to make it ultra-compact.
3. Simulate in HFSS/ADS, fabricate on standard FR-4, and measure it on a VNA.

- **The Publishable Angle:** "Design of a Compact Defected Ground Structure (DGS) Diplexer for Wi-Fi/Bluetooth Coexistence in Dense IoT Nodes."
- **Target Venues:** *IEEE Access*, *MDPI Electronics*, or *IEEE MTT-S Latin America / Regional Microwave Conferences*.

Project 6: Conformal/Wearable Antenna for 24 GHz FMCW Radar (Vital Signs)

- **The Concept:** 24 GHz and 60 GHz FMCW radars are increasingly used for non-contact vital sign monitoring (breathing/heart rate) in smart homes. However, when a radar antenna is placed on a curved surface or near the human body, the detuning effect ruins the radar link budget.
- **The Task:**
  1. Design a 2-element patch antenna array for 24 GHz FMCW radar in CST/HFSS.
  2. Simulate the antenna on a standard human body phantom (available in all modern EM software) to observe the detuning and efficiency drop.
  3. Propose a simple matching network or ground-plane modification to recover the antenna's performance when bent or placed near tissue.
- **The Publishable Angle:** "Analysis and Mitigation of Human-Body Detuning Effects on 24 GHz Wearable FMCW Radar Antennas for Vital Sign Monitoring."
- **Target Venues:** *IEEE Sensors Journal*, *IEEE Radio Wireless Week (Student Track)*, or *IEEE AP-S Symposium*.

Project 7: 5G/Wi-Fi 6E Rectenna with Harmonic Rejection for Ambient Backscatter

- **The Concept:** "Zero-power" IoT devices harvest ambient RF energy (from Wi-Fi routers or 5G towers) to power themselves. The core component is a "Rectenna" (Antenna + Rectifier diode). A major issue is that the non-linear diode generates harmonics that get re-radiated, causing EMI.
- **The Task:**
  1. Design a simple 2.4 GHz or 5.8 GHz patch antenna.
  2. Design an impedance matching network between the antenna and a Schottky diode (using ADS harmonic balance simulation).
  3. Add a compact low-pass or bandstop filter at the DC output to prevent harmonics from re-radiating (an EMC fix).
  4. Measure the RF-to-DC conversion efficiency vs. input power in the lab.
- **The Publishable Angle:** "A Compact 5.8 GHz Rectenna with Integrated Harmonic Suppression for Low-Power Ambient IoT Backscatter Applications."
- **Target Venues:** *IEEE Antennas and Wireless Propagation Letters (AWPL)*, *IEEE International Conference on RFID*, or *IEEE Microwave and Wireless Technology Letters (MWTL)*.

Project 8: GPS Receiver Front-End Susceptibility to 5G/LTE Out-of-Band Blockers

- **The Concept:** GPS signals are incredibly weak (around -130 dBm). A nearby 5G cell tower or LTE base station transmitting at high power can easily saturate a GPS Low Noise Amplifier (LNA), even if the 5G signal is "out of band," due to LNA non-linearity.
- **The Task:**
    1. Model a standard commercial GPS LNA (e.g., using a vendor S-parameter/NL model in ADS).
    2. Inject a strong out-of-band "blocker" signal (e.g., a 700 MHz LTE signal) into the simulation.
    3. Calculate the desensitization (how much the GPS noise figure degrades).
    4. Design and simulate a simple, low-loss passive pre-filter to protect the LNA and prove it restores the GPS sensitivity.
- **The Publishable Angle:** "System-Level Analysis and Mitigation of LTE/5G Out-of-Band Desensitization in High-Sensitivity GPS RF Front-Ends."
- **Target Venues:** *IEEE International Symposium on Electromagnetic Compatibility (EMC)*, *IEEE Transactions on Electromagnetic Compatibility*, or *ION GNSS+ Conference*.

Project Execution in a 16-Week Timeline

- **Week 3: Literature Review & Abstract (10% of grade)**
    - Students must read 10 recent IEEE papers on their chosen topic and write a 250-word abstract mimicking an IEEE submission.
- **Week 6: Introduction & System Model (Midterm 1 Deliverable)**
    - Students submit the first two sections of their IEEE paper (Introduction + System Architecture).
- **Week 10: Simulation Results & Draft (Mid-Project Review)**
    - Students must have their ADS/HFSS/MATLAB simulations finished. They present their raw data (graphs, Smith charts, eye diagrams) in class.
- **Week 12: Draft Paper Submission (Midterm 2 Deliverable)**
    - Students submit a complete, 4-page draft of the IEEE paper (Results, Discussion, Conclusion).
- **Week 15: Final Presentation & Revisions**
    - Students present their work. Peer and instructor feedback is used to polish the paper.
- **Week 16: Final Camera-Ready Paper**
    - The final deliverable is a formatted IEEE conference/letter paper ready to be uploaded to IEEE Xplore or submitted to a journal.